

35-Day Advanced Industry Readiness Program

DSA + Backend Development + Generative AI for Developers

Duration: 35 Days | Daily Effort: 4–5 Hours

Target Audience: Advanced B.Tech CSE Students preparing for Product Companies, AI Engineering Roles, and Full-Stack Development Careers.

Program Outcomes

Students will master advanced Data Structures & Algorithms, Backend Development, System Design, API Engineering, Databases, Cloud-Native Development, LLM Integration, RAG Pipelines, AI Agents, and GenAI-powered Software Engineering practices.

35-Day Delivery Plan

Day	Topic	Duration	Hands-on Deliverable
Day 1	Complexity Analysis, Competitive Programming Environment, STL Deep Dive	4-5 Hours	Lab + Coding Assignment
Day 2	Advanced Arrays, Prefix/Suffix Techniques, Sliding Window	4-5 Hours	Lab + Coding Assignment
Day 3	Binary Search Patterns, Parametric Search	4-5 Hours	Lab + Coding Assignment
Day 4	Linked Lists, Fast-Slow Pointer Techniques	4-5 Hours	Lab + Coding Assignment
Day 5	Stacks, Queues, Monotonic Stack Applications	4-5 Hours	Lab + Coding Assignment
Day 6	Trees Fundamentals, Traversals, Binary Trees	4-5 Hours	Lab + Coding Assignment
Day 7	BST, AVL, Segment Trees Introduction	4-5 Hours	Lab + Coding Assignment
Day 8	Heap, Priority Queue, Top-K Problems	4-5 Hours	Lab + Coding Assignment
Day 9	Hashing, Maps, Sets, String Hashing	4-5 Hours	Lab + Coding Assignment

Day 10	Graphs I – BFS, DFS, Connected Components	4-5 Hours	Lab + Coding Assignment
Day 11	Graphs II – Shortest Paths, Dijkstra	4-5 Hours	Lab + Coding Assignment
Day 12	Graphs III – MST, Union Find, DSU	4-5 Hours	Lab + Coding Assignment
Day 13	Graphs IV – Topological Sort, SCC	4-5 Hours	Lab + Coding Assignment
Day 14	Dynamic Programming I	4-5 Hours	Lab + Coding Assignment
Day 15	Dynamic Programming II – Advanced Patterns	4-5 Hours	Lab + Coding Assignment
Day 16	Greedy Algorithms and Optimization	4-5 Hours	Lab + Coding Assignment
Day 17	Tries, Bit Manipulation, Advanced Problems	4-5 Hours	Lab + Coding Assignment
Day 18	Mock Assessment + Coding Contest	4-5 Hours	Lab + Coding Assignment
Day 19	Backend Architecture, REST Principles, Node.js Runtime	4-5 Hours	Lab + Coding Assignment
Day 20	Express.js Advanced Routing and Middleware	4-5 Hours	Lab + Coding Assignment
Day 21	Authentication, JWT, OAuth2, RBAC	4-5 Hours	Lab + Coding Assignment
Day 22	MongoDB Advanced Data Modeling	4-5 Hours	Lab + Coding Assignment
Day 23	SQL Optimization, Indexing, Query Tuning	4-5 Hours	Lab + Coding Assignment
Day 24	Microservices Architecture	4-5 Hours	Lab + Coding Assignment
Day 25	API Gateway, Event Driven Systems	4-5 Hours	Lab + Coding Assignment
Day 26	Docker, Containers, CI/CD Basics	4-5 Hours	Lab + Coding Assignment
Day 27	Redis Caching, Message Queues	4-5 Hours	Lab + Coding Assignment
Day 28	System Design for Scalable Applications	4-5 Hours	Lab + Coding Assignment
Day 29	Introduction to LLMs,	4-5 Hours	Lab + Coding Assignment

	Transformers, Prompt Engineering		
Day 30	OpenAI APIs, Gemini APIs, AI SDK Integration	4-5 Hours	Lab + Coding Assignment
Day 31	Embeddings, Vector Databases, Semantic Search	4-5 Hours	Lab + Coding Assignment
Day 32	RAG Architecture and Knowledge Retrieval	4-5 Hours	Lab + Coding Assignment
Day 33	AI Agents, Tool Calling, MCP Concepts	4-5 Hours	Lab + Coding Assignment
Day 34	GenAI Product Development Project	4-5 Hours	Lab + Coding Assignment
Day 35	Capstone Demo, Interview Preparation, Hiring Readiness	4-5 Hours	Lab + Coding Assignment

Major Projects

1. Competitive Programming Portfolio (DSA)
2. Enterprise Backend API Platform
3. Scalable Microservices Application
4. RAG-based Enterprise Knowledge Assistant
5. AI Agent for Software Development Automation
6. Final Industry-Grade Capstone

Assessment Strategy

- Daily Coding Challenges – 20%
- Weekly Assessments – 20%
- Backend Project Evaluation – 20%
- GenAI Project Evaluation – 20%
- Final Capstone & Viva – 20%

Technology Stack

DSA: C++, STL, LeetCode, Codeforces

Backend: Node.js, Express.js, MongoDB, PostgreSQL, Redis, Docker

Cloud & DevOps: GitHub, CI/CD, Docker

GenAI: OpenAI, Gemini, LangChain, Vector Databases, RAG, AI Agents